Plan B

De-specializing SALMONN-SQA into a calibrated, multi-dimensional speech-quality rater

Turning a lenient noise specialist into a model that perceives, names, and scores every common degradation — and emits a de-compressed, calibrated MOS.

Every training input is public — LibriTTS-R + MUSAN + OpenSLR SLR28 RIRs.

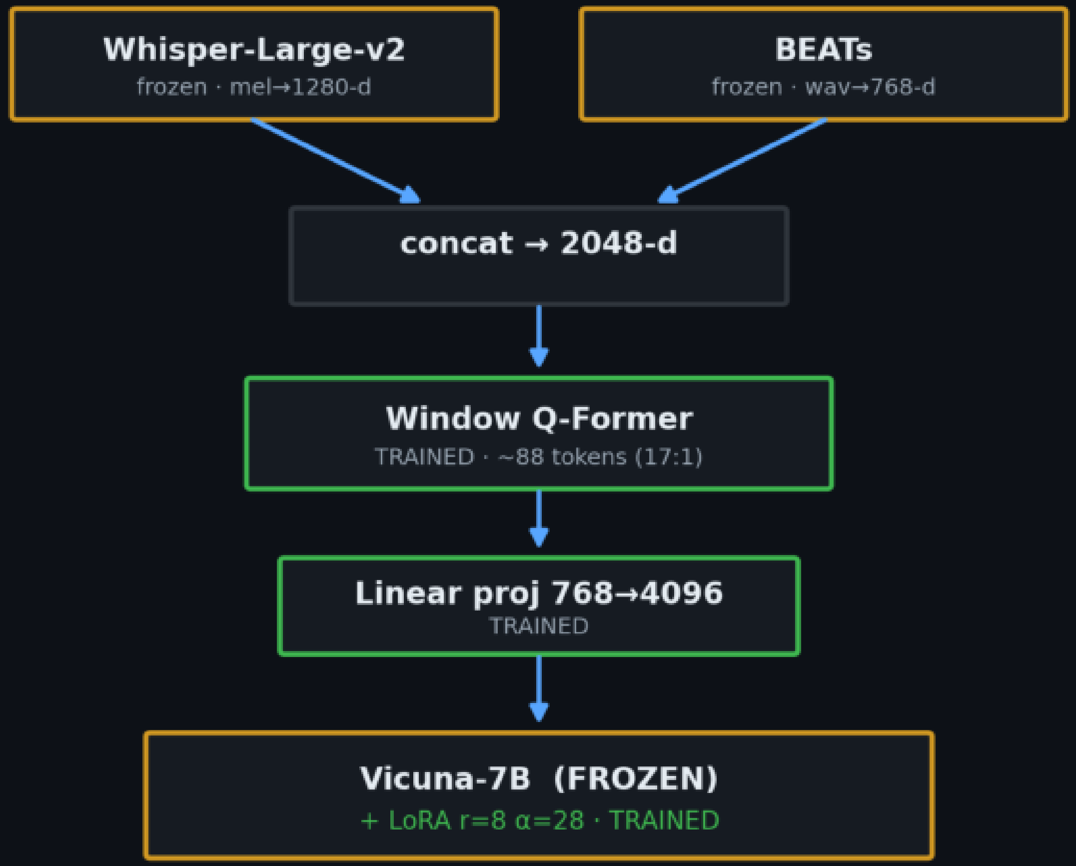
Clement Laroche · published model ckpt_stage2_open · VoiceBank-DEMAND held-out

The model: SALMONN-7B, SQA-finetuned

Audio LLM: two frozen encoders → a trained audio bottleneck → frozen Vicuna-7B + small LoRA.

- Whisper-Large-v2 — mel→1280-d. ASR-trained, deliberately reverb-invariant.
- BEATs — raw waveform → 768-d acoustic-event features.
- Window Q-Former — ~0.33s windows → ~88 tokens. TRAINED. ~17:1 compression.
- Linear projection 768→4096, spliced at <SpeechHere>. TRAINED.
- Vicuna-7B-v1.5 + LoRA (r=8, α=28). Base FROZEN, LoRA TRAINED.

Plan B trains only the green blocks + LoRA. Encoders stay frozen.



The problem: a lenient noise specialist

Good additive-noise descriptions — but as a quality meter, weak and blind to whole classes.

① MOS isn't a usable scale

- Coarse 5 discrete values, floored at 2.5. Agrees with PESQ/NISQA/DNSMOS at only ρ 0.40–0.49
  - — yet those metrics agree with each other at 0.72–0.82. It is the outlier.

② Blind to whole degradation types

- Reverb ρ(MOS,severity) = −0.11 (no response). Weak on bandwidth (−0.39), clipping (−0.33).

③ Blind to enhancement

- Denoise: every objective metric improves, MOS moves +0.03.

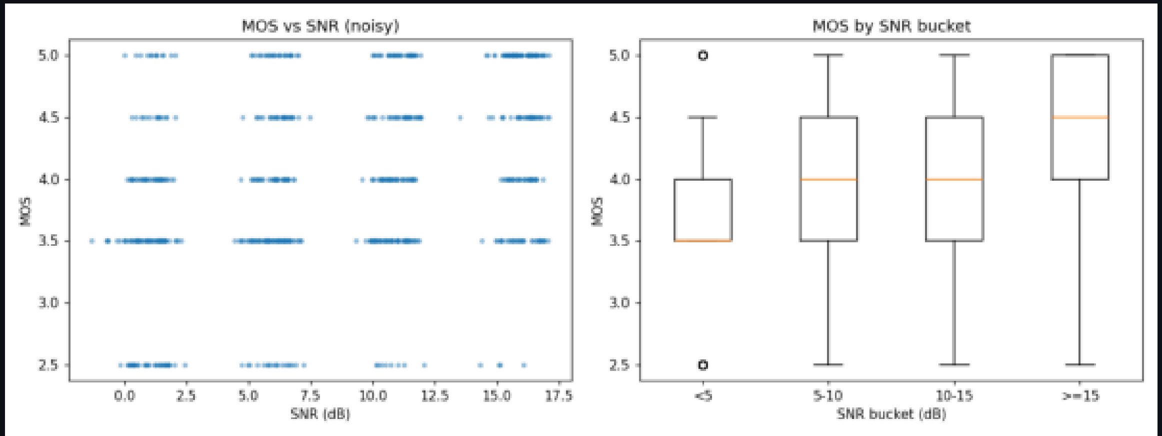
④ Brittle output

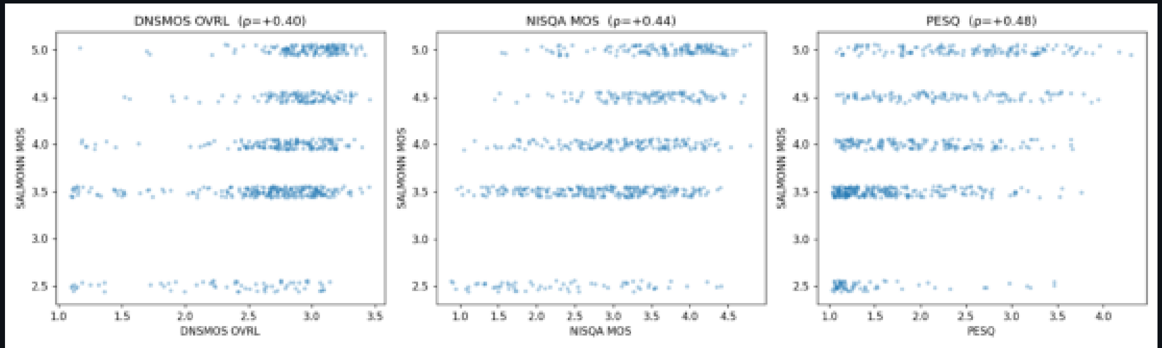
- OOD degradation → made-up JSON schema. Acquiescent: leading yes/no → always YES.

Goal: keep the descriptive ability; add perception + calibration across ALL degradations.

Evidence of the blindness (original model)

- Left: MOS barely tracks SNR, and sits in a narrow high band — the floored, compressed scale.
- Right: vs DNSMOS / NISQA / PESQ the MOS quantizes into a few horizontal stripes (ρ 0.40–0.49).





Experiment design: de-risk, then train

Ask the key question cheaply, first

- Whisper is reverb-invariant; Q-Former compresses 17:1. If reverb/bandwidth info is destroyed
  - in the frozen front-end, no LoRA training recovers it → would need to unfreeze encoders (heavy).

The hypothesis

- Blindness lives in the LLM's learned read-out (only ever trained on noise), not the architecture.

If true, the fix is data + targets — not surgery on the encoders.

Stage 0 — is degradation decodable from frozen features?

Known-parameter degradations → features at 4 frozen taps → linear probe (PCA+ridge),

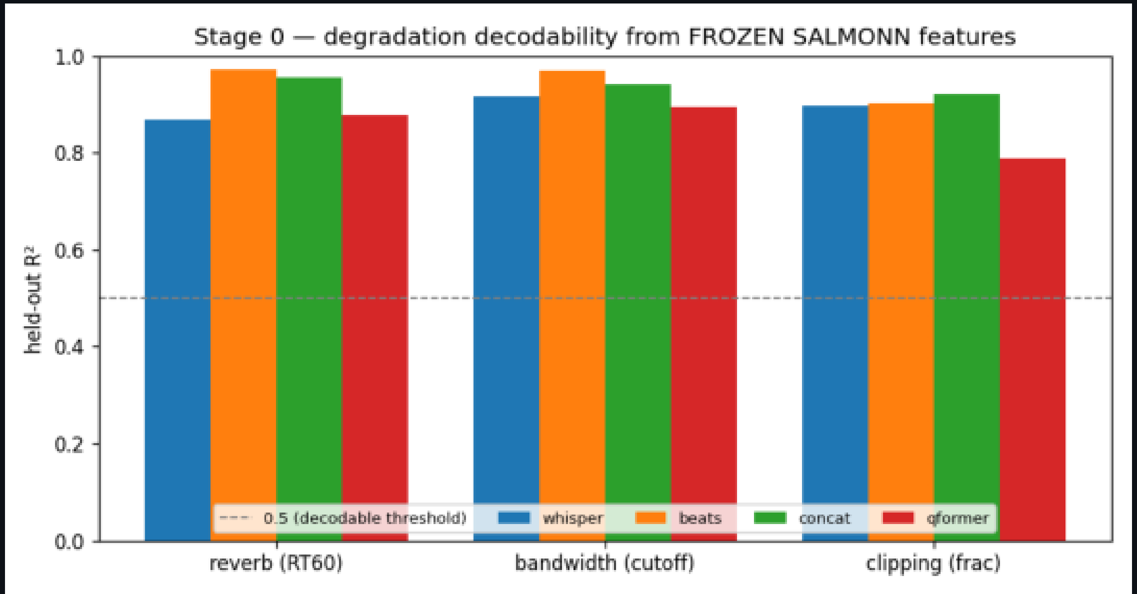
scored by R^2+Spearman, held out by utterance.

A linear probe is the point: tests whether info is linearly decodable, not buried.

Taps: whisper · beats · concat · qformer (the bottleneck that reaches the LLM)

Result — PASSED. R^2 0.82–0.95, Spearman 0.90–0.98 at every tap, incl. Q-Former output.

- Decision: keep encoders frozen; the fix is data + targets.



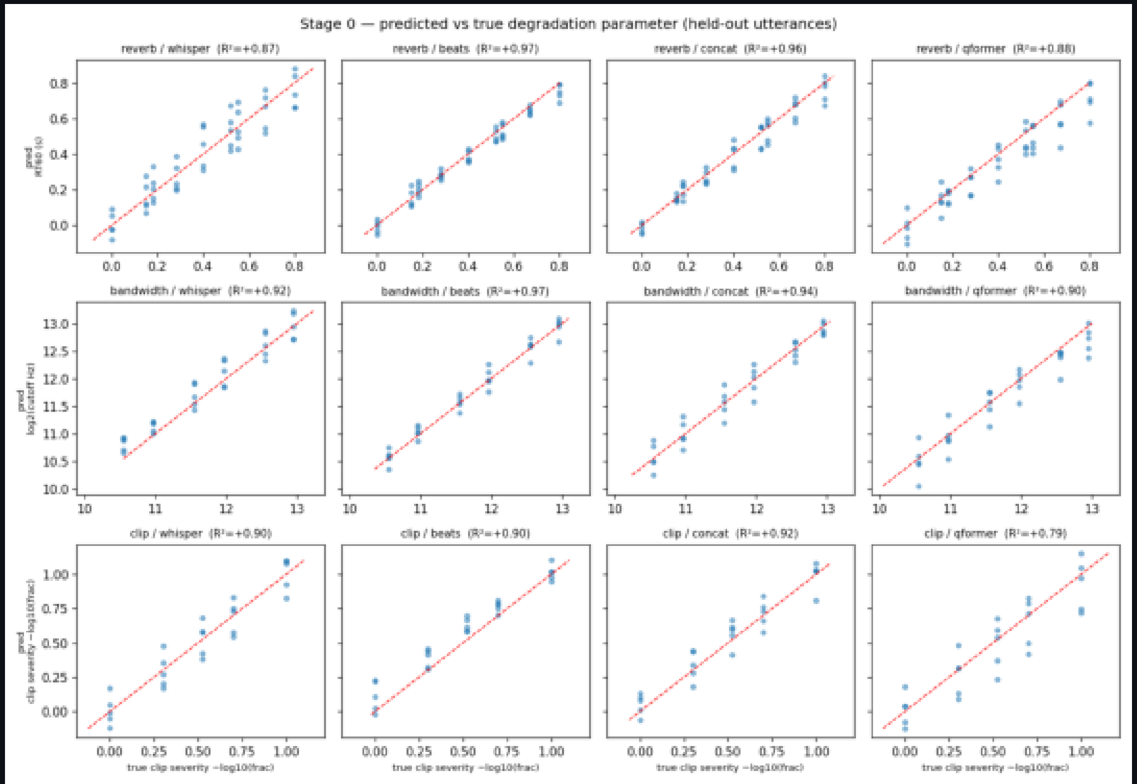
Stage 0 — the signal survives to the LLM's doorstep

Where the blindness actually is

- Predicted-vs-true RT60 / cutoff / clip-fraction sit on y=x at every tap, incl. qformer.
- The degradation reaches the LLM intact — the model was never taught to read it
  - (its read-out head was trained only on additive noise).

Each axis is probed one-at-a-time → proves per-axis detectability, not type-discrimination.

Discriminating WHICH degradation is the residual that the synthetic-type data fixes.



Data generation: on-the-fly, known-parameter corpus

Clean speech — leakage-safe

- Train: LibriTTS-R train-clean-100 — 4,000 clips. Val: dev-clean — 300 clips.
- No VCTK speakers → disjoint from VoiceBank-DEMAND eval. No DEMAND noise in training.

Description text — Opus 4.8 paraphrase

- ~740 unique profiles → one Opus 4.8 call each → grounding-verified prose
- All 4,300 descriptions paraphrased, 0 template fallbacks.

axis	how	train
reverb	SLR28 simulated RIRs (Apache-2.0) + synthetic	2,138
noise	real MUSAN + synth white/pink/brown	512
codec	real Opus/MP3 → bandwidth	509
bandwidth	Butterworth low-pass	DSP
clipping / disc.	hard-clip / frame-drop	DSP
loudness	re-gain	DSP
clean	untouched	373

Every dataset is openly licensed (CC BY 4.0 / Apache-2.0), so the whole corpus regenerates from public data.

Choice of artefacts & the severity map

8 degradation classes, each scored 1–5 from its exact applied parameter (5=pristine,1=severe).

Three design choices

- Perceptual bands — thresholds follow speech-quality convention (telephone BW ≈ 3.4 kHz).
- Reverb floors at 4 for any applied RIR — measured RIRs measured at distance, never truly dry;
  - very low DRR (<−15 dB) drops one more band. (Fixed an early RT60-0.15-scored-clean mislabel.)
- Use the measured effect, not the knob — clip/disc. on measured fraction; codec→measured rolloff.

axis	parameter → score bands
noise	SNR: ≥30→5, 20-30→4, 12-20→3, 6-12→2, <6→1
reverb	RT60: <.25→4, .45→3, .70→2, >.70→1; DRR<−15 → −1
bandwidth	cutoff: ≥7500→5, 6000→4, 4000→3, 2500→2, <2500→1
clipping	frac: 0→5, <1%→4, 5%→3, 15%→2, >15%→1
discontinuity	loss: 0→5, <2%→4, 5%→3, 10%→2, >10%→1
loudness	|Δgain|: ≤3→5, 6→4, 12→3, 20→2, >20→1

Targets: calibrate-then-describe

Per-dimension 1–5 scores

- From exact params — direct fix for 'quality prior overrides the named dimension'.

Overall MOS

- 0.55·min + 0.45·mean of dim scores (worst-axis-dominant), blended 70/30 with fused
  - PESQ+NISQA+DNSMOS. Pure metric fusion rejected: PESQ floors on reverb, DNSMOS reverb-blind.

Expected I/O

- Input: waveform + sqa_full instruction.
- Stage 1 target: score block ONLY (learn calibrated numbers first).
- Stage 2 target: scores + paraphrased description + Overall MOS.

Cross-entropy on target tokens only; prompt + 88 audio embeds masked to −100.

Training: two-stage LoRA-SFT

What LoRA does here

- Learn tiny low-rank adapters (r=8, α=28) in Vicuna's projections — a few M params that
  - re-shape the read-out without disturbing the 7B frozen base.
- Trainable: Q-Former + speech→LLaMA proj + Vicuna-LoRA. Frozen: Whisper, BEATs, Vicuna base.

single GPU · AMP · batch 2 × accum 8 · lr 1e-5 · warmup→cosine

Why two stages

- Learn calibrated numbers in isolation first (no description tokens competing for gradient),
  - then describe + emit MOS on top of a representation that already rates correctly.

stage	target text	from	ep
1 — calibration / (sqa_score)	per-dimension score block only	released SQA ckpt	2
2 — reasoning / (sqa_full)	scores + description + Overall MOS	Stage 1 best	4

Results: all five findings re-measured

Held-out synthetic degradation sweep on VoiceBank-DEMAND clean (disjoint speakers).

#	Original finding	published (open)
1	MOS↔SNR ρ 0.37, lenient	ρ 0.46; penalizes low SNR; names noise 98% (low SNR)
2	ρ 0.40–0.49; 5 values floored	ρ 0.65–0.71; 63 distinct values (1.90–4.89)
3	reverb −0.27, bw −0.37, clip −0.48	reverb −0.80, bw −0.88, clip −0.81, noise −0.93
4	Enhancement MOS +0.03, ρ≈0	MOS gain +1.05 (91% of files), ρ +0.32
5	Acquiescence → always YES	calibrated discriminative scores; N/A

5→63
distinct MOS values

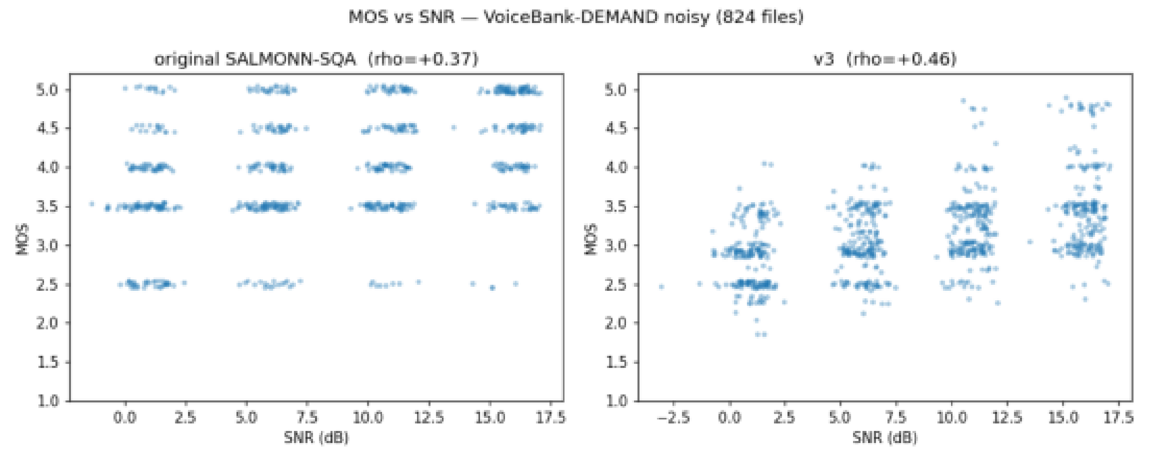
+0.83
measured-reverb ρ vs PESQ

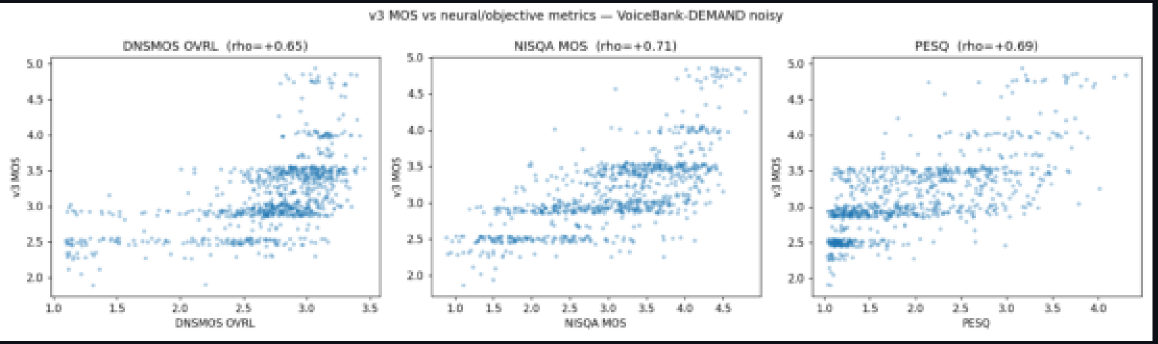
0 / 108
degenerate outputs

+1.05
enhancement MOS gain

Results: the MOS scale is rebuilt

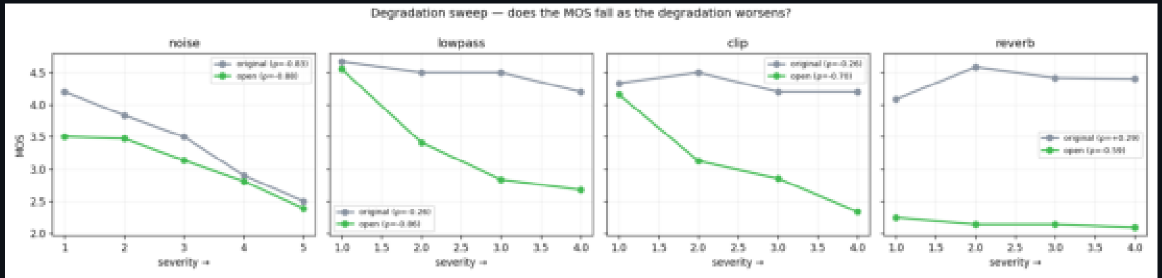
- Left: MOS vs SNR — the cloud spreads the full range and slopes with SNR (was a flat band).
- Right: MOS vs metrics — continuous, rank-aligned (ρ 0.65–0.71); the stripes are gone.

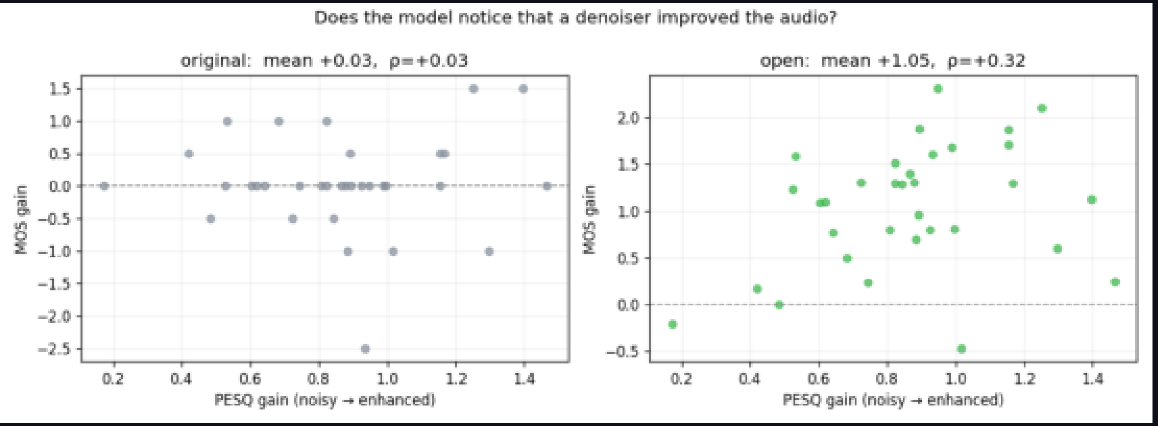




Results: every axis now responds

- Left: MOS vs severity — monotone downward everywhere, incl. the former reverb blind spot.
- Right: enhancement — MOS rises with denoiser gain (+1.05, 91% of files): a usable enhancer evaluator.



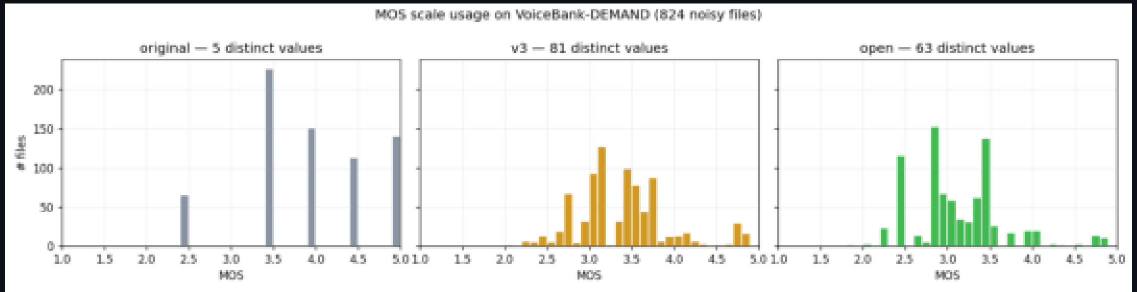


Results: from a 5-rung ladder to a continuous meter

- Original emitted a handful of MOS values, floored well above the bottom of the scale.

The published model uses 63 distinct values spanning 1.90–4.89 — it meters quality, not buckets.

This de-compression lets rank-correlations climb into the band where the metrics agree.



The reverb result: the benchmark was measuring an artefact

Scored against PESQ — which knows nothing about either severity map

- synth_reverb injects an explicit direct path → our synthetic reverb has an

  – artificially HIGH DRR at every RT60. v3 keys on RT60 alone — exactly how that

  – sweep grades severity. So the sweep flattered v3 and penalized open.

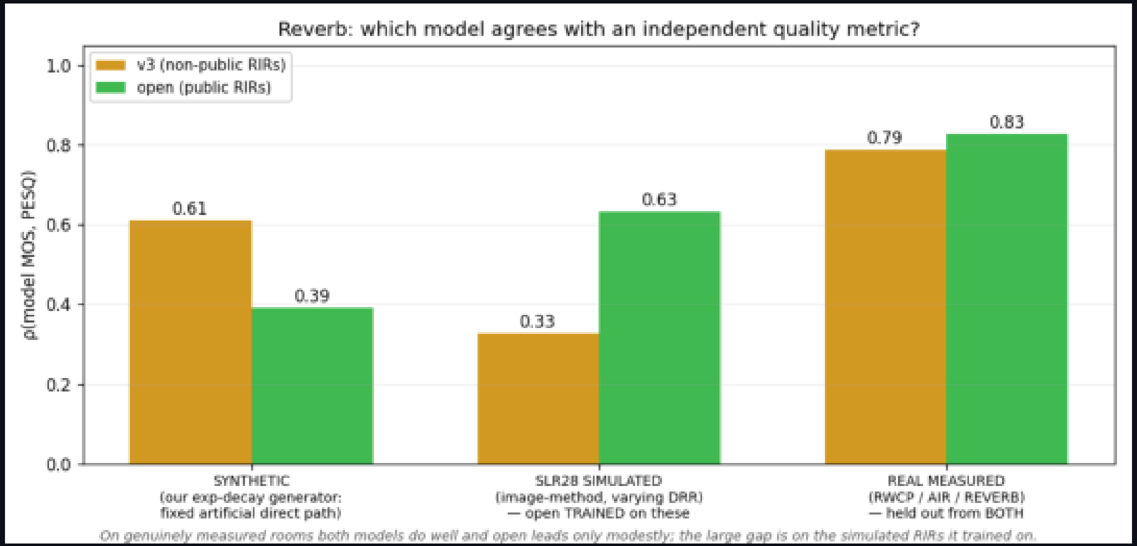
On REAL measured rooms both models do well; open leads modestly (0.83 vs 0.79).

The big gap is on SLR28 SIMULATED RIRs — which open trained on. That is an

– in-distribution advantage, not generalization. Reporting it as 'real rooms'

– would have been an overclaim (an earlier draft of this control made exactly that

– mistake: the pool is 60k simulated vs 320 measured



Bottom line

The fine-tune turns a lenient noise specialist with a weak 5-value MOS into a calibrated rater:

- MOS de-compressed (5→63), rank-aligned with PESQ/NISQA/DNSMOS (ρ 0.65–0.71)
- Perceives and ranks every degradation type — incl. the former reverb blind spot
- Tracks enhancement gain (+1.05 MOS, 91% of files) → a usable automatic enhancer evaluator
- Names degradations in natural, grounded prose; 0 degenerate outputs

Per-dimension structured scores are the most reliable signal. The whole fix was data + targets —

encoders never unfrozen, validated up-front by the Stage 0 probe.

Optional further work

- Scale clean pool to train-clean-360 · more degradation types · scalar regression head if banding remains.